Falcon 9 landing outcome analysis

Historical course data, reproducible queries and model comparisons

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23 October 2026

Executive summary

The 90-record API/reference cohort contains 60 positive landing labels (66.7%). The label includes successful ocean outcomes and must not be treated as a direct measure of recoverable booster reuse.

Logistic regression scores 15/18 on the held-out set. This small historical test does not establish future flight reliability or a launch-cost forecast.

Separate 101-record SQL and 56-record map/dashboard cohorts answer their own questions. Keeping their denominators distinct prevents misleading comparisons.

Introduction

Question: which recorded launch characteristics are associated with the course landing label, and how well can simple classifiers predict it within this historical dataset?

The workflow combines data preparation, exploratory plots, ten SQL queries, an interactive Folium map, a Dash dashboard and a held-out classification comparison.

This is an educational observational analysis. Mission success, a successful landing and physical recovery for reuse are different outcomes.

Data collection and provenance

API collection: normalize the retained course JSON snapshot, restrict to single-core/single-payload Falcon 9 records through 13 November 2020, and cross-check the resulting 90 launches against the published reference CSV.

The live metadata endpoint returned non-JSON in the earlier collection. The recorded launch dates and reference data identify the Falcon 9 rocket ID. The analysis discloses this fallback rather than claiming a new live API extraction.

SQL uses the supplied 101-record Spacex.csv. Mapping and dashboard use their supplied 56-record snapshots. These cohorts have different naming conventions and coverage.

Wrangling and label definition

Replace missing PayloadMass values with the cohort mean and verify that none remain. Keep the 26 missing landing-pad values. The retained get_dummies default gives them an all-zero indicator block, which needs review before deployment.

Class = 1 when Outcome starts with True; otherwise Class = 0. This yields 41 True ASDS + 14 True RTLS + 5 True Ocean = 60 positive labels.

One-hot encode Orbit, LaunchSite, LandingPad and Serial. The resulting modelling matrix has 90 rows and 80 columns. Identify variables that could leak post-launch information before any deployment.

Exploratory and interactive methods

Compare flight number and payload mass with launch site and orbit. Show both binary outcomes and group sizes, then examine the annual trend.

Execute all ten specified SQL tasks against the retained SQLite table and retain the query text next to its result.

Folium provides clustered outcome markers and a coastline-reference distance. Dash filters launch site and payload range and updates the success pie and payload scatter views.

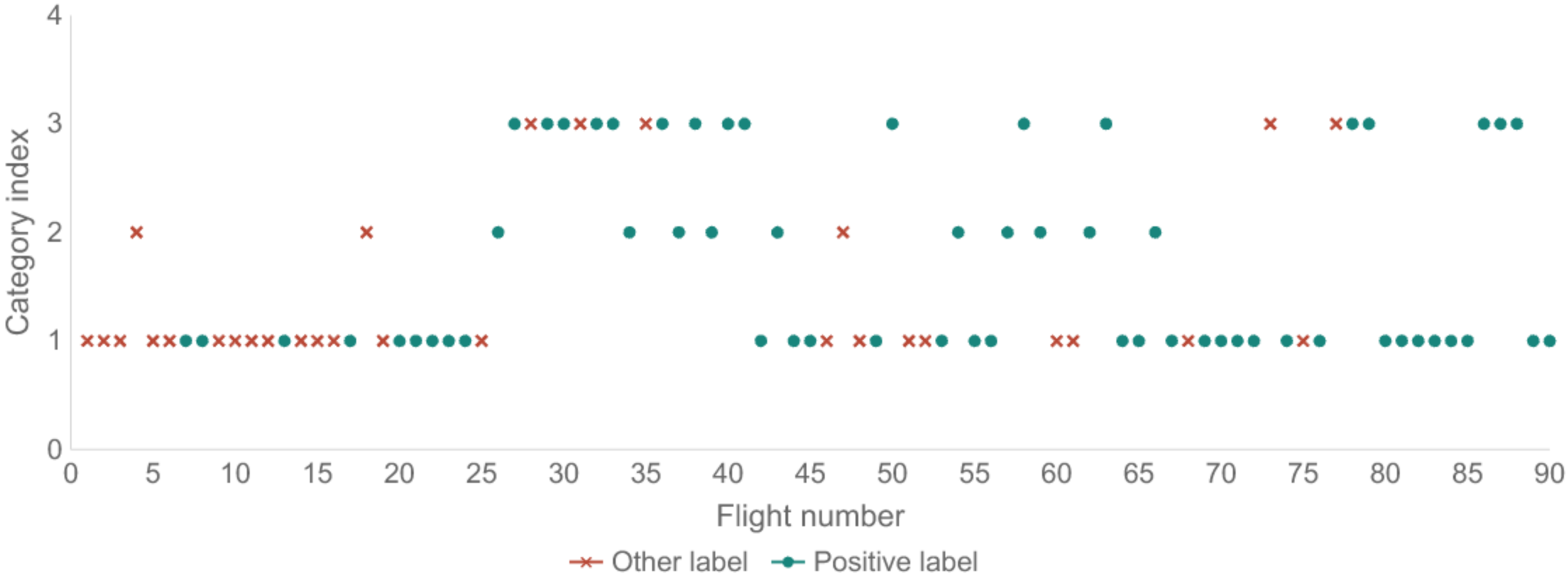
Predictive analysis method

Split 90 records into 72 training and 18 test records with random_state = 2. Use five-fold GridSearchCV on the training set and compare logistic regression, SVM, decision tree and KNN.

StandardScaler sits inside each estimator pipeline, so each cross-validation fit learns scaling from its own training fold. The same split and scoring metric support the comparison.

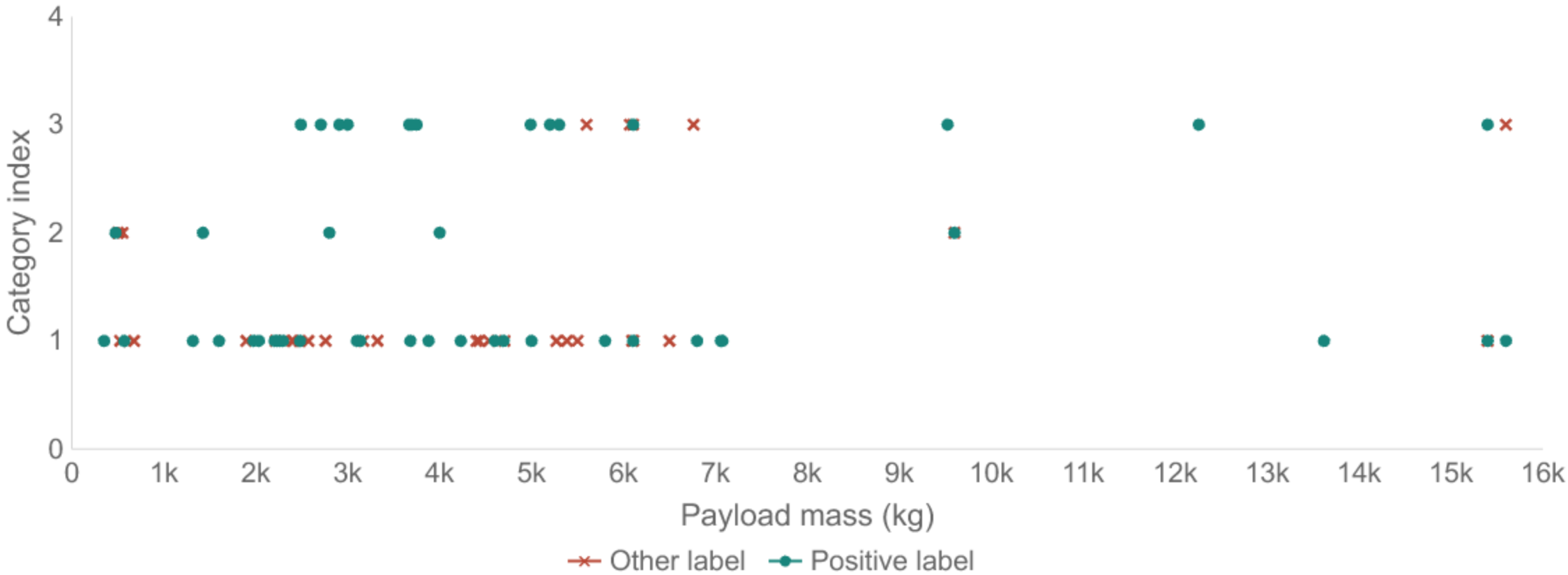
The prior course-reference experiment uses a different 83-feature matrix and preprocessing path. Its scores are kept separate from the 80-feature pipeline results shown here.

Flight number and launch site



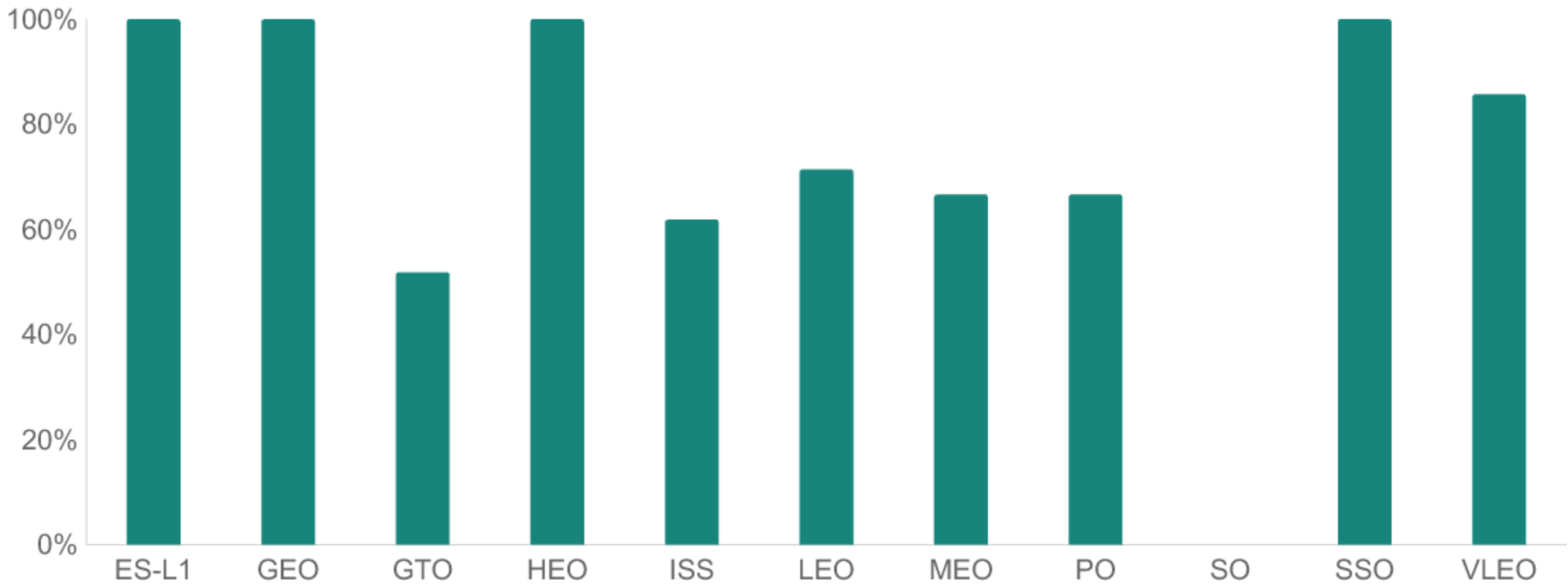
Later launches contain more positive labels. Site and time are confounded. 1=CCAFS SLC 40; 2=VAFB SLC 4E; 3=KSC LC 39A

Payload mass and launch site



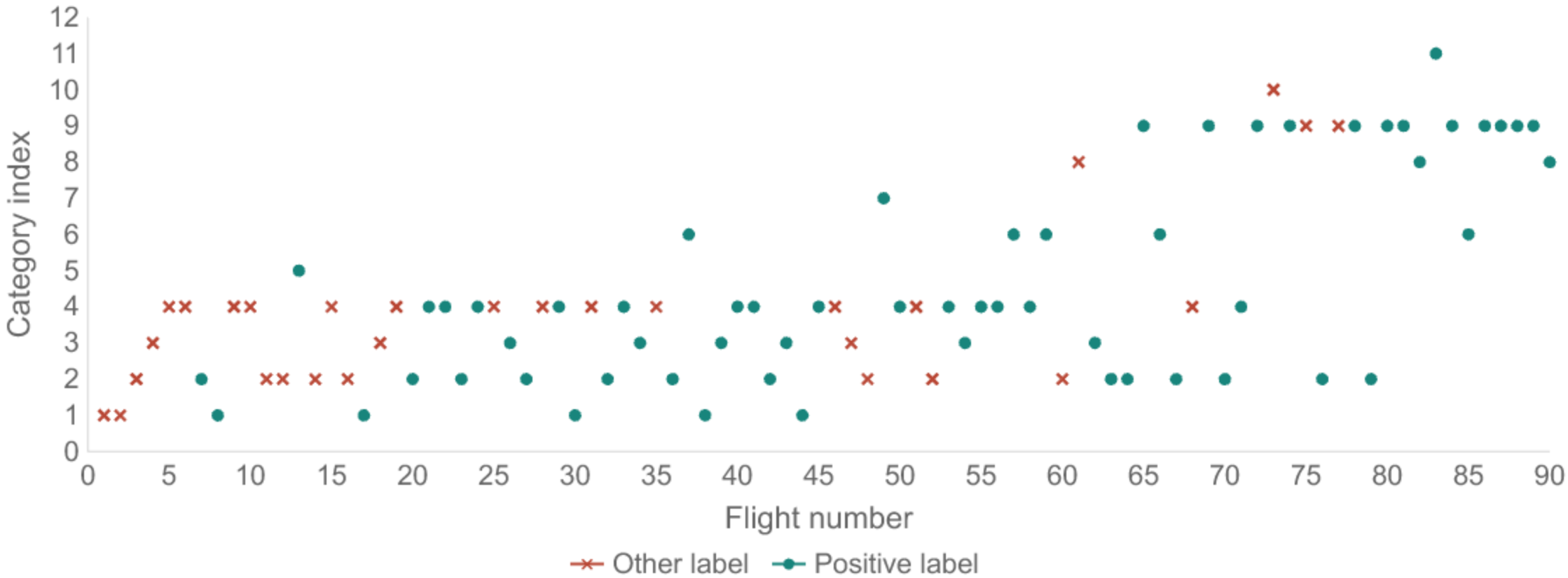
Both labels occur across payload masses; the plot alone supplies no causal payload threshold. 1=CCAFS SLC 40; 2=VAFB SLC 4E; 3=KSC LC 39A

Landing-label rate by orbit



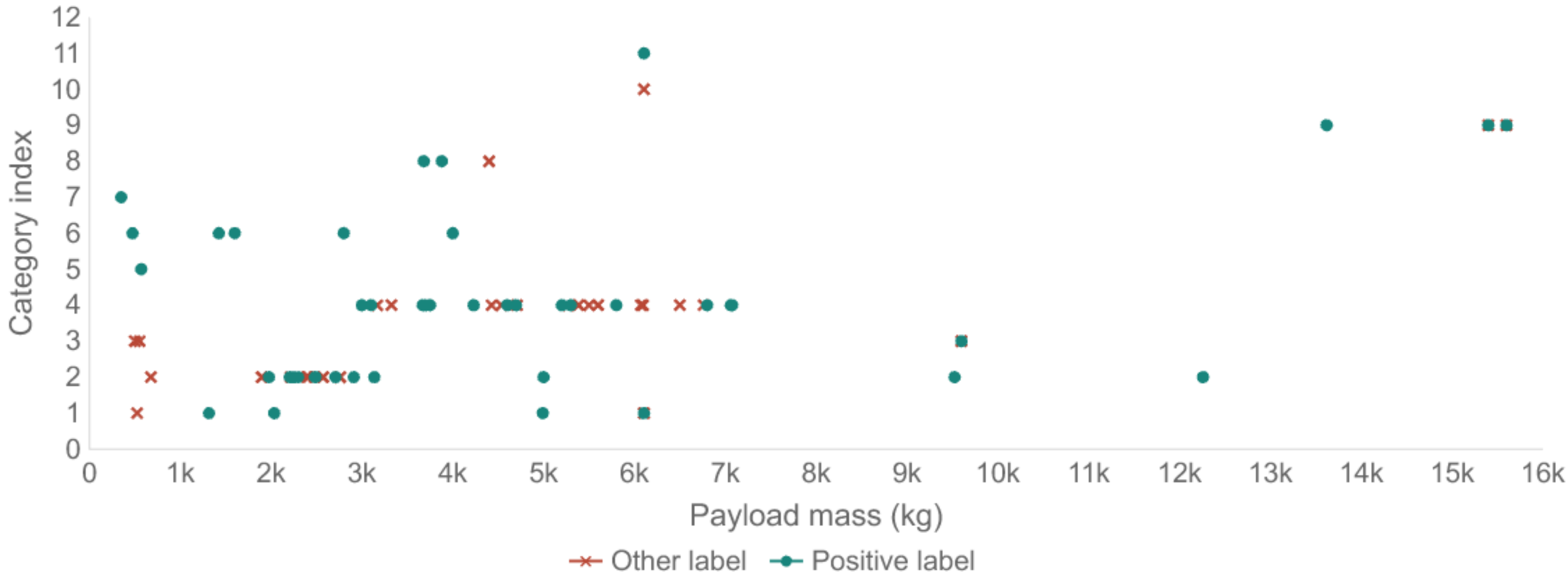
Several orbit categories have only one observation. Rates from those groups are highly uncertain.

Flight number and orbit



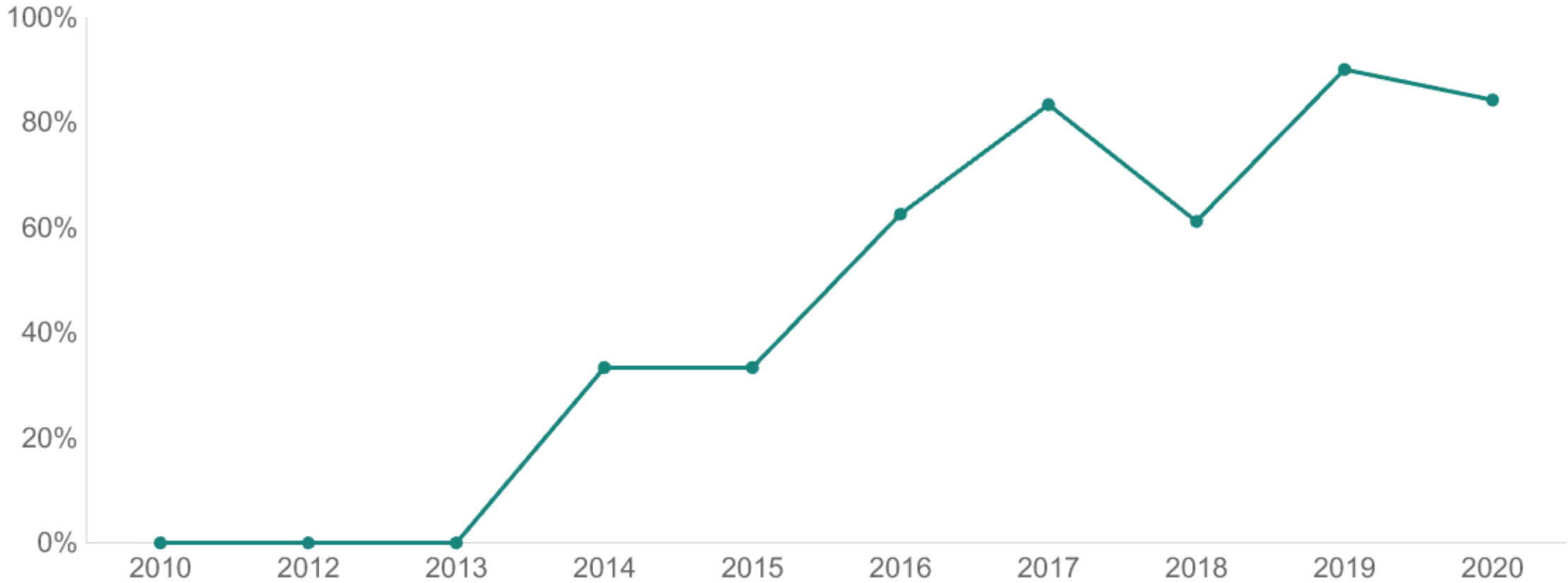
The orbit mix changes over time, so flight-number associations also reflect mission selection. 1=LEO; 2=ISS; 3=PO; 4=GTO; 5=ES-L1; 6=SSO; 7=HEO; 8=MEO; 9=VLEO; 10=SO; 11=GEO

Payload mass and orbit



Payload and orbit jointly describe missions; neither association establishes causation. 1=LEO; 2=ISS; 3=PO; 4=GTO; 5=ES-L1; 6=SSO; 7=HEO; 8=MEO; 9=VLEO; 10=SO; 11=GEO

Annual landing-label rate



Historical 90-record cohort: 2017 15/18; 2018 11/18; 2019 9/10; 2020 16/19. Small early cohorts limit comparisons.

SQL 1: Unique launch sites

```
SELECT DISTINCT Launch_Site FROM SPACEXTBL ORDER BY Launch_Site
```

Launch_Site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 2: Five CCA launch records

SELECT Date, Booster_Version, Launch_Site, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5

Date	Booster_Version	Launch_Site	Payload (kg)
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	0
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	0
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	525
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	500
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	677

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 3: NASA CRS payload mass

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_kg FROM SPACEXTBL WHERE Customer='NASA (CRS)'
```

Total_kg
45596

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 4: F9 v1.1 average payload

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Average_kg FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1%'
```

Average_kg
2534.667

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 5: First successful ground landing

```
SELECT MIN(Date) AS First_date FROM SPACEXTBL WHERE Landing_Outcome='Success (ground pad)'
```

First_date
2015-12-22

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 6: Drone-ship boosters, 4000-6000 kg

```
SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome='Success (drone ship)' AND
PAYLOAD_MASS__KG_>4000 AND PAYLOAD_MASS__KG_<6000 ORDER BY Booster_Version
```

Booster_Version
F9 FT B1021.2
F9 FT B1031.2
F9 FT B1022
F9 FT B1026

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 7: Mission outcomes

```
SELECT TRIM(Mission_Outcome) AS Mission_Outcome, COUNT(*) AS Missions FROM SPACEXTBL GROUP BY
TRIM(Mission_Outcome) ORDER BY Missions DESC
```

Mission_Outcome	Missions
Success	99
Success (payload status unclear)	1
Failure (in flight)	1

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 8: Boosters with maximum payload (1/2)

```
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_=(SELECT
MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL) ORDER BY Booster_Version
```

Booster_Version	Payload (kg)
F9 B5 B1048.4	15600
F9 B5 B1048.5	15600
F9 B5 B1049.4	15600
F9 B5 B1049.5	15600
F9 B5 B1049.7	15600
F9 B5 B1051.3	15600

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 8: Boosters with maximum payload (2/2)

```
SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_=(SELECT
MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL) ORDER BY Booster_Version
```

Booster_Version	Payload (kg)
F9 B5 B1051.4	15600
F9 B5 B1051.6	15600
F9 B5 B1056.4	15600
F9 B5 B1058.3	15600
F9 B5 B1060.2	15600
F9 B5 B1060.3	15600

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 9: Drone-ship landing failures in 2015

```
SELECT substr(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE substr(Date,1,4)='2015' AND Landing_Outcome='Failure (drone ship)' ORDER BY Date
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 10: Landing outcomes within date interval (1/2)

SELECT Landing_Outcome, COUNT(*) AS Outcomes FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcomes DESC

Landing_Outcome	Outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

SQL 10: Landing outcomes within date interval (2/2)

SELECT Landing_Outcome, COUNT(*) AS Outcomes FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcomes DESC

Landing_Outcome	Outcomes
Failure (parachute)	2
Precluded (drone ship)	1

Source: supplied 101-record SQL cohort. Payload units are kg; query predicates and date bounds are explicit.

Folium launch-site map

Site	Latitude	Longitude
CCAFS LC-40	28.562302	-80.577356
CCAFS SLC-40	28.563197	-80.576820
KSC LC-39A	28.573255	-80.646895
VAFB SLC-4E	34.632834	-120.610746

The generated launch_map.html contains four site markers and a 56-launch outcome cluster. Browser policy prevented opening local HTML here; no live-map screenshot is claimed.

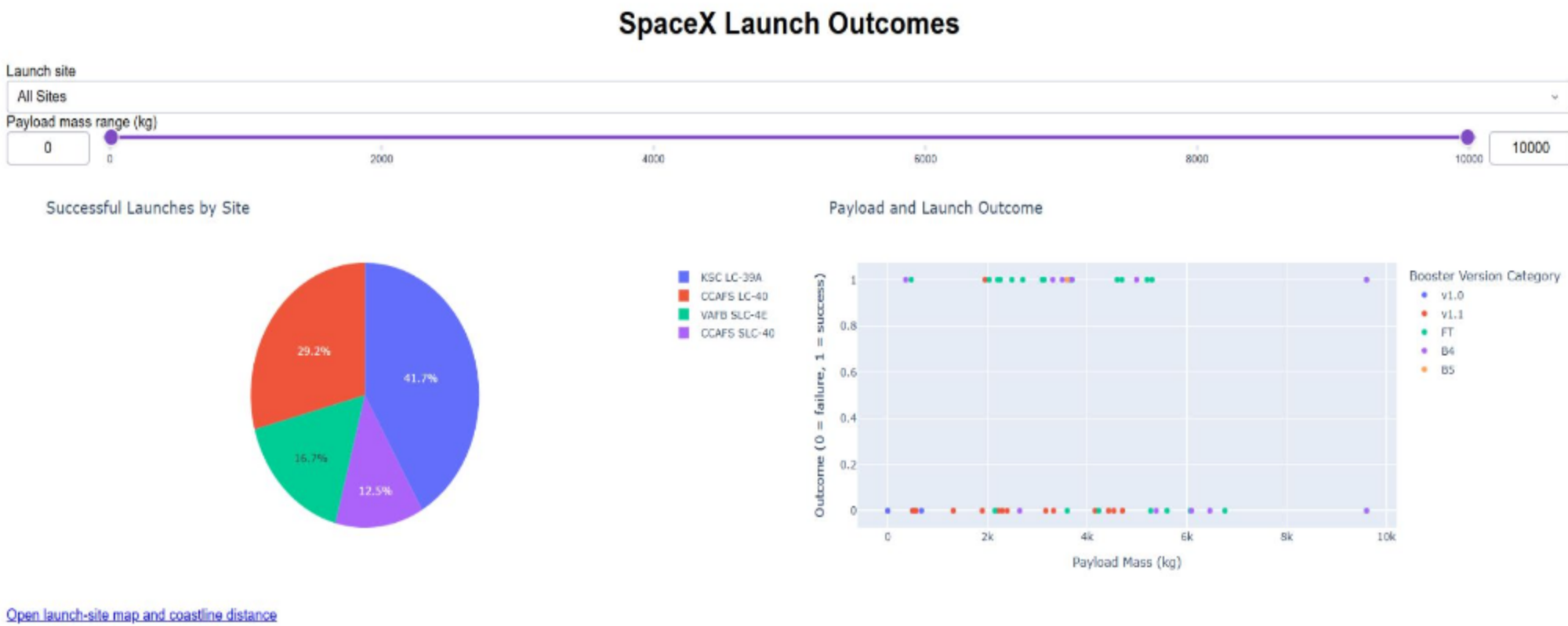
Folium outcome and proximity results

The retained map includes a green/red marker for each of 56 launch labels, a MarkerCluster layer and mouse-position coordinates. Overlapping markers at the same site require clustering for inspection.

Using the supplied coastline point (28.56367, -80.57163) and the haversine formula gives 0.579758 km from CCAFS LC-40. This is distance to that reference point, not a surveyed minimum coastline distance.

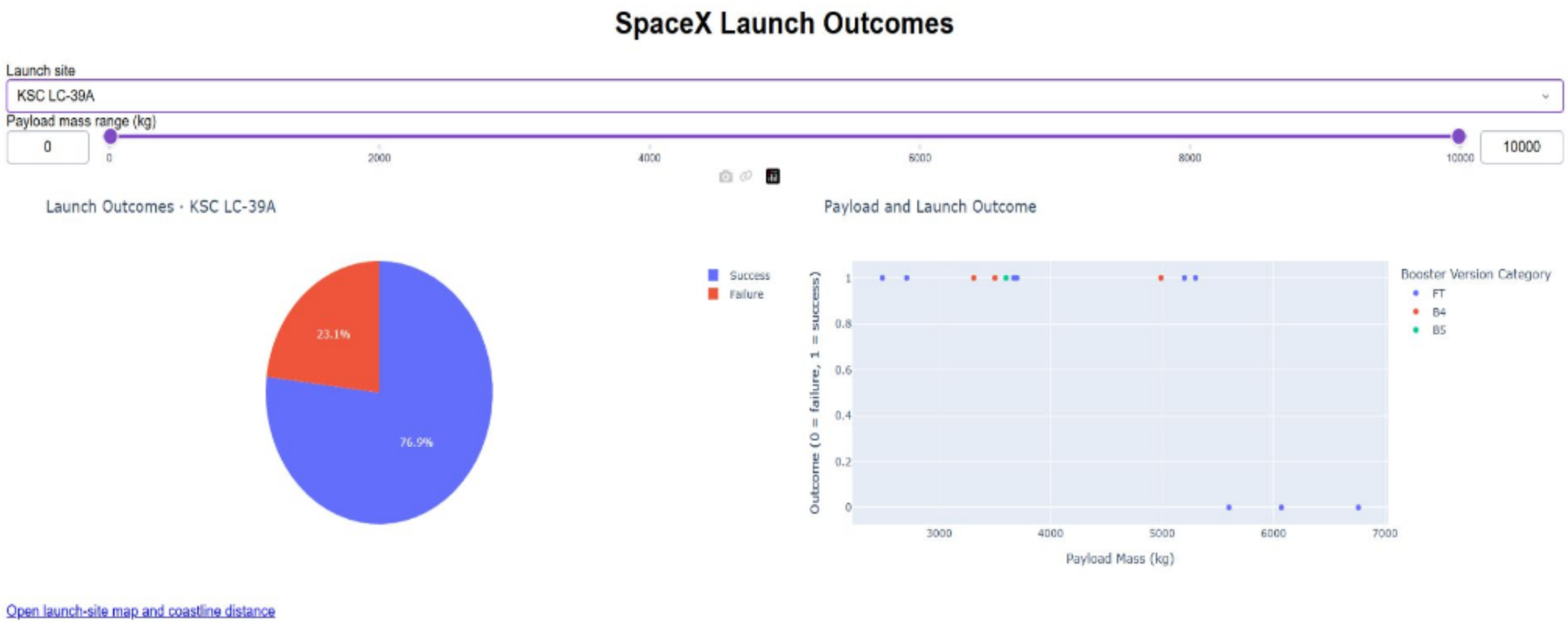
The map is a descriptive geographic view. It cannot isolate geography from launch date, mission type or hardware evolution.

Dash dashboard: all sites



Retained screenshot of the implemented Dash app. The complete Python callback source accompanies the project.

Dash dashboard: filtered view



Callback checks cover ALL, each of four sites, and the 2000-5000 kg payload range. All scatter values stay within the requested filter.

Dashboard cohort: success counts and denominators

Launch site	Positive	Records	Fraction
CCAFS LC-40	7	26	26.9%
CCAFS SLC-40	3	7	42.9%
KSC LC-39A	10	13	76.9%
VAFB SLC-4E	4	10	40.0%

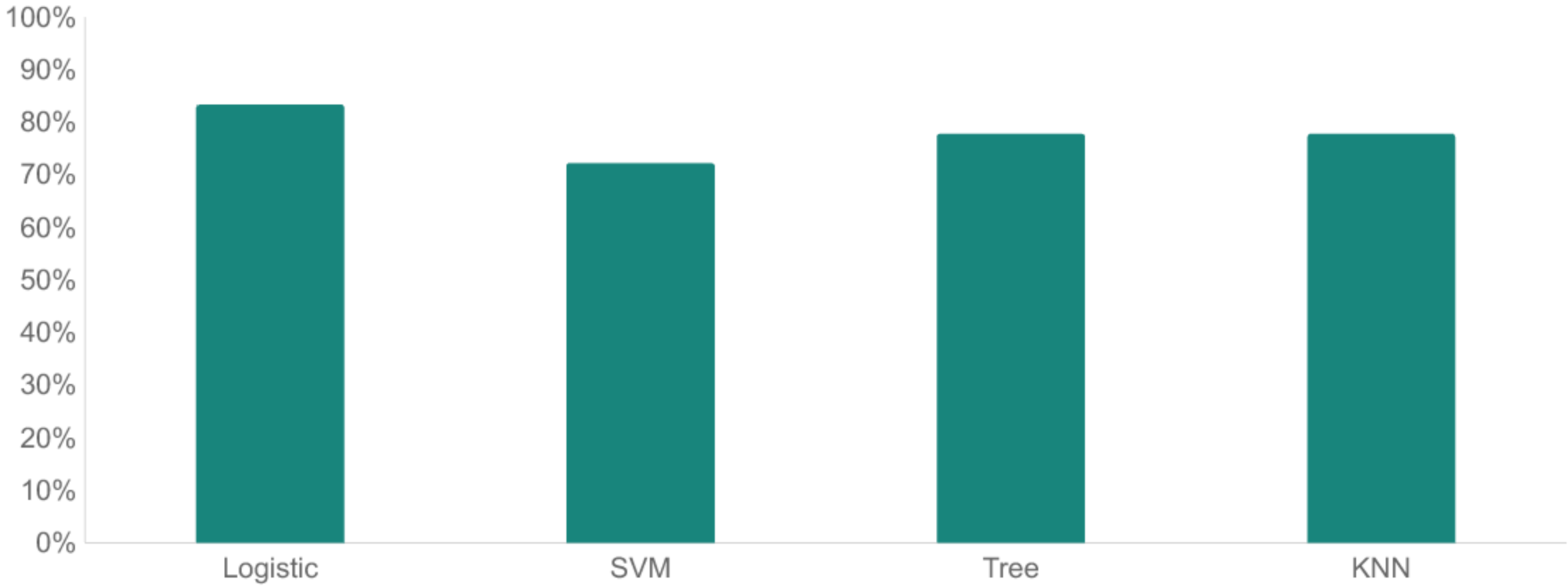
56-record dashboard cohort. These fractions are not the API cohort site rates; the two CCAFS labels remain distinct in this supplied dataset.

Classifier comparison

Model	Training CV	Held-out accuracy	Correct / 18
Logistic Regression	80.76%	83.33%	15/18
SVM	80.67%	72.22%	13/18
Decision Tree	83.24%	77.78%	14/18
KNN	73.52%	77.78%	14/18

Five-fold CV on training data; separate held-out test. The highest CV score and highest test score belong to different models.

Held-out accuracy



Logistic regression: 15/18. A one-record change shifts accuracy by 5.6 percentage points. Avoid overinterpreting this small test set.

Logistic Regression: confusion matrix

True class	Predicted 0	Predicted 1
0	3	3
1	0	12

Actual held-out predictions from the 80-feature pipeline. A false positive predicts a positive landing label when the recorded label is 0.

SVM: confusion matrix

True class	Predicted 0	Predicted 1
0	3	3
1	2	10

Actual held-out predictions from the 80-feature pipeline. A false positive predicts a positive landing label when the recorded label is 0.

Decision Tree: confusion matrix

True class	Predicted 0	Predicted 1
0	3	3
1	1	11

Actual held-out predictions from the 80-feature pipeline. A false positive predicts a positive landing label when the recorded label is 0.

KNN: confusion matrix

True class	Predicted 0	Predicted 1
0	2	4
1	0	12

Actual held-out predictions from the 80-feature pipeline. A false positive predicts a positive landing label when the recorded label is 0.

Selected parameters and validation limits

Best training parameters: logistic C=1; SVM C=10 with RBF kernel; decision-tree maximum depth=8; KNN k=3. Model selection uses training cross-validation, not repeated tuning against the held-out labels.

Temporal drift, high-cardinality serial categories and potentially post-event variables limit deployment relevance. A future experiment should define pre-launch-only features and a chronological holdout.

Do not use the reported accuracy to infer a dollar saving or an operational flight-safety probability. The label and sampling process do not support those conclusions.

Additional insight: outcome and cohort definitions

Five positive ocean outcomes contribute to the 60/90 course label. A reuse-focused question should distinguish controlled landing, ocean contact, physical recovery and later reuse before defining its target.

Three supplied cohorts have 90, 101 and 56 records. Keeping these populations separate explains why site counts and success fractions differ without inventing a data-quality correction.

A defensible follow-up would reconcile launch identifiers, document intended deployment time, and evaluate a pre-launch-only chronological model with calibrated probabilities.

Conclusion

The project provides six exploratory visualizations, ten executed SQL queries, map and dashboard analysis, and a reproducible four-model comparison on historical course data.

Logistic regression achieved the strongest held-out score in this experiment, while the decision tree led cross-validation. The test set is too small to claim a reliable model ranking for future missions.

The generated map and complete source files support further inspection. A useful next step is a chronological evaluation with pre-launch-only variables and an outcome definition tied to physical recovery.

Sources and reproducibility

Course data and notebooks: IBM Applied Data Science Capstone, including retained API_call_spacex_api.json, dataset_part_1/2.csv, Spacex.csv and the map/dashboard CSV snapshots.

Original analysis: checks.py, eda_checks.py, remaining_checks.py, interactive_dashboard.py and ten additional SQL queries. JSON verification files retain counts, filters, parameters and scores.

The public launch API metadata fallback and blocked local-map browser preview are explicit. The reproducibility bundle retains the cohort data, queries, Python source and executed notebooks.